

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	Substitution: Moment = 2.0×50 [not: 2.0×0.5] (1) = 100 [N cm] (1) Do not accept 1.0 N m	1	1		2	2	2
		(ii)	Substitution: Distance = $100 \text{ (ecf)} \div 2.5$ (1) = 40 [cm] (1)	1	1		2	2	2
		(iii)	[For the moments to balance because] $1.5 \text{ N} < 2.0 \text{ N}$ [or the force on the right < the force on the left] (1).... the distance [from the pivot] must be greater than 50 cm / beyond the end of the ruler [which is impossible]. (1)			2	2		2
	(b)		Substitution: Weight = $15\,000 \times 10$ (1) = 150 000 [N] (1)	1	1		2	2	2
			Question 5 total	3	3	2	8	6	8